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Even Semester Mid Term Examination, March 2025
Faculty of Science, Technology and Architecture, SCSE/SE
Department of IT/IOT&IS/CCF/DSE/BCE/CE/ME/Mechatronics
B.Tech.

Course Code: MAS2001 Course: Statistical Methods & Probability Theory
Time: 1.5 hrs.

Semester: IV
Max. Marks: 30

Instructions: All questions are compulsory.
Missing data, if any, may be assumed suitably.
Calculator is allowed.

SECTION A

- S. No. Marks
- A1 If T_1 and T_2 are two unbiased estimators based on the random sample $X_1, X_2, \dots, X_n$. Then T_2 is said to be efficient if [2]
(a) $\text{Var}(T_1) < \text{Var}(T_2)$
(b) $\text{Var}(T_1) = \text{Var}(T_2)$
(c) $\text{Var}(T_1) > \text{Var}(T_2)$
(d) $\text{Var}(T_1) = \text{Var}(T_2) = 1$
- A2 The mean and variance of a binomial distribution are 10 and 5, respectively. Find $P(X=0)$ [2]
(a) $\frac{1}{2^5}$ (b) $\frac{1}{2^8}$ (c) $\frac{1}{2^{10}}$ (d) $\frac{1}{2^{20}}$
- A3 Let X be a random variable with $E[X]=5$ and $E[X^2]=30$. What is the variance of the random variable $Y=X-4$? [2]
(a) 5 (b) 6 (c) 10 (d) 11

SECTION B

- B1 A random sample $(X_1, X_2, X_3, X_4, X_5)$ of size 5 is drawn from a normal population with unknown mean μ . Consider the following estimators to estimate μ : [4]
(i) $t_1 = \frac{X_1 + X_2 + X_3 + X_4 + X_5}{5}$, (ii) $t_2 = \frac{X_1 + X_2}{2} + X_3$, (iii) $t_3 = \frac{2X_1 + X_2 + \beta X_3}{3}$.
(a) Are t_1 and t_2 unbiased?
(b) Find the value of β such that t_3 is an unbiased estimator of μ .
(c) Which is the best estimator among t_1, t_2 and t_3 .
- B2 A random variable X has the following probability function: [4]
- | Values of $X, x:$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|-------------------|-----|------|------|------|-------|--------|------------|
| $P(x):$ | C | $2C$ | $2C$ | $3C$ | C^2 | $2C^2$ | $7C^2 + C$ |
- (a) find C
(b) evaluate $P(X < 6)$, $P(X \geq 6)$, and $P(1 \leq X < 5)$
(c) determine the distribution function of X
(d) if $P(X \leq a) > \frac{1}{2}$, find the minimum value of a .

B3 Show that in estimating the parameter μ in a Poisson distribution with probability mass function $\frac{e^{-\mu} \mu^x}{x!}$, $\bar{x}$ is sufficient for μ . [4]

B4 If the Chebyshev's inequality for the random variable X is given by $P(-2 < X < 8) \geq \frac{21}{25}$, find $E(X)$ and $Var(X)$. [4]

SECTION-C

C1 In an intelligence test administered on 1000 children, the average was 60 and standard deviation was 20. Assuming that the marks obtained by the children follow a normal distribution, find the number of children who have scored (a) over 90 marks (b) below 40 marks and (c) between 50 and 80 marks (d) what is the probability that in a group of 5 children 2 have scored marks between 50 and 80 marks. [8]
Useful data: $\phi(1.5)=0.4332$, $\phi(1)=0.3413$, $\phi(0.5)=0.1915$,

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